DATA EXPLORATION -EXCEL

Module -1 - Excel

Assignment Task Overview

Problem Statement

You are a Data Analyst, and you are required to perform basic data exploration and analysis in

Microsoft Excel. The dataset contains information about different products, including their Product

IDs, Product Name, Brand Name, Quantity, Categories, and Prices.

In this assignment, you will use Excel formulas and functions to perform calculations, categorize

data, and extract useful information from text fields. The goal is to build foundational skills in data

exploration, logical functions, conditional aggregation, and text manipulation using Excel.

The focus of this assignment is on data summarization, conditional logic, and text extraction

techniques, which are essential for preparing data for further analysis

Dataset Description

Dataset: Product Dataset

Attributes: Product ID, Product Name, Brand Name, Quantity, Category, Price

Tasks:

DATA EXPLORATION -EXCEL

Basic Data Exploration

3. Logical Function – IF

● Create a new column named Price Range. 1. Sum, Count, and Average

Perform the following calculations using Excel functions:

● Calculate the total price of all products in the dataset.

● Determine how many products exist in the dataset.

● Calculate the average price of the products.

2. Minimum and Maximum Values

Using Excel functions:

● Identify the minimum price among all products.

● Identify the maximum price among all products.

● Use the IF function to categorize products based on their price:

○ If Price ≥ $500 → classify as High Price

○ If Price < $500 → classify as Standard Price

4. Conditional Functions – SUMIF and COUNTIF

Perform the following conditional calculations:

● Use the SUMIF function to calculate the total price of products in the

Electronics category.

● Use the COUNTIF function to determine the number of products with a price

less than $100.

5. Text Formatting – LEFT, RIGHT, MID

● Use text functions to extract information from the Product ID column.

● Create the following new columns:

○ Create a new column named Day with the first 2 characters of each

'Product ID' using the LEFT function.

○ Create a new column named Country Code by extracting the last 2

characters from the 'Product ID' column using the RIGHT function.

○ Create a new column named Month by extracting the 4th to 6th

characters from the 'Product ID' column using the.